

```

[INFO] T E S T S
[INFO] -----
[INFO] Running data_management.PatientTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.024 s -- in data_management.PatientTest
[INFO] Running data_management.DataStorageTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in data_management.DataStorageTest
[INFO] Running com.alerts.PatternIntegrationTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.006 s -- in com.alerts.PatternIntegrationTest
[INFO] Running com.alerts.AlertGeneratorTest
ALERT TRIGGERED: Patient 2 - Condition: Low Oxygen Saturation: 89.0% at time 2000
ALERT TRIGGERED: Patient 1 - Condition: URGENT: Critical Blood Pressure: 190.0 at time 1000
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.004 s -- in com.alerts.AlertGeneratorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 11, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.368 s
[INFO] Finished at: 2026-05-10T18:49:57+02:00
[INFO]

```

In Phase 4, the system was refactored to use the **Singleton pattern** for centralized data management. The **Factory Method** and **Strategy patterns** were integrated to decouple alert creation from monitoring logic, making it easier to add new medical metrics in the future. Finally, the **Decorator pattern** was implemented to allow for dynamic modification of alert properties, such as priority levels, without altering the base alert classes.